

Skip Module Hyperparameter Selection

1 Algorithm

Algorithm 1 Skip Module Hyperparameter Selection

Require: Parameter space Θ , validation set D_{val} , DL model m , recall tolerance δ_R , overhead ratio β , weights w_S, w_R

Ensure: Optimal parameters θ^*

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1: baseline  $\leftarrow$  EVALUATEPIPELINE( $D_{\text{val}}$ , skip=None,  $m$ )
2: Extract  $R_{\text{base}}, T_{\text{DL}}$  from baseline
3: best_score  $\leftarrow -\infty$ ;  $\theta^* \leftarrow \text{None}$ 
4: for each  $\theta \in \Theta$  do
5:   results  $\leftarrow$  EVALUATEPIPELINE( $D_{\text{val}}, \mathcal{S}(\theta), m$ )
6:   Extract  $R(\theta), S_r(\theta), T_{\text{skip}}(\theta)$  from results
7:   if  $R(\theta) \geq R_{\text{base}} - \delta_R$  and  $T_{\text{skip}}(\theta) \leq \beta \cdot T_{\text{DL}}$  then
8:      $\rho(\theta) \leftarrow 1 - \frac{R_{\text{base}} - R(\theta)}{\delta_R}$ 
9:     score  $\leftarrow w_S S_r(\theta) + w_R \rho(\theta)$ 
10:    if score > best_score then
11:      best_score  $\leftarrow$  score
12:       $\theta^* \leftarrow \theta$ 
13:    end if
14:  end if
15: end for
16: return  $\theta^*$ 
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The skip module $\mathcal{S}$ contains several hyperparameters (e.g., motion threshold, accumulation window size) that can significantly impact the performance of the overall system. To identify the optimal configuration, we employ a grid search based hyperparameter optimization procedure on a validation set D_{val} derived from our video dataset with respect to several system-wide metrics defined in Section ???. More specifically, to select the optimal hyperparameters for the proposed skip module, we employ a constrained optimization procedure on the validation set D_{val} . Let R_{base} denote the baseline recall, i.e., the fraction of fire/smoke frames correctly detected by the pipeline without skipping, and let T_{DL} denote its mean per-frame inference time. For each candidate parameter set $\theta \in \Theta$ obtained by grid search, we evaluate the full pipeline $\text{Read} \rightarrow \mathcal{S}(\theta) \rightarrow [m]$ and compute three metrics: recall $R(\theta)$ (the fraction of fire/smoke frames

correctly detected), skip rate $S_r(\theta)$ (the fraction of ground-truth negative frames skipped by $\mathcal{S}$), and mean per-frame skip module time $T_{\text{skip}}(\theta)$. Formal metric definitions are given in Section ??.

The skip rate $S_r(\theta)$ is computed on D_{val} following the definition in Section ??, with N_{neg} instantiated as $N_{\text{neg}}^{\text{val}} = |\{f_i \in D_{\text{val}} : y_i = 0\}|$, which is fixed by the validation set and independent of θ .

The mean per-frame inference times are defined as:

$$T_{\text{DL}} = \frac{1}{|D_{\text{val}}|} \sum_{i=1}^{|D_{\text{val}}|} t_{\text{DL}}(f_i), \quad (1)$$

$$T_{\text{skip}}(\theta) = \frac{1}{|D_{\text{val}}|} \sum_{i=1}^{|D_{\text{val}}|} t_{\text{skip}}(f_i, \theta), \quad (2)$$

where f_i denotes the i -th frame in D_{val} , and both averages are taken over all frames to reflect the runtime cost on a typical video stream.

Feasibility constraints. A candidate θ is considered feasible only if it satisfies two hard constraints:

$$R(\theta) \geq R_{\text{base}} - \delta_R, \quad (3)$$

$$T_{\text{skip}}(\theta) \leq \beta \cdot T_{\text{DL}}, \quad (4)$$

where recall tolerance $\delta_R > 0$ is the maximum allowable absolute recall drop, and overhead ratio $\beta \in (0, 1)$ bounds the skip module overhead as a fraction of one full DL inference. For example, setting $\delta_R = 0.01$ and $\beta = 0.10$ means the system tolerates at most a 1% recall drop and requires the skip module to run within 10% of the cost of a single DL inference.

Scoring feasible candidates. Among feasible candidates, both objectives are normalized over Θ_{feasible} before weighting, so that the weights w_S and w_R reflect the intended relative importance rather than the raw scale of each term. The range-normalized recall retention and skip rate are defined as:

$$\tilde{\rho}(\theta) = \frac{\rho(\theta) - \min_{\Theta_f} \rho}{\max_{\Theta_f} \rho - \min_{\Theta_f} \rho}, \quad \tilde{S}_r(\theta) = \frac{S_r(\theta) - \min_{\Theta_f} S_r}{\max_{\Theta_f} S_r - \min_{\Theta_f} S_r}, \quad (5)$$

where $\Theta_f \equiv \Theta_{\text{feasible}}$, and $\rho(\theta)$ is defined as in Equation (??). Both normalized terms are bounded in $[0, 1]$ over the feasible region, ensuring that the weights w_S and w_R directly control the relative influence of each objective on the final ranking.

Optimal selection. The optimal parameter set θ^* is selected as

$$\theta^* = \arg \max_{\theta \in \Theta_{\text{feasible}}} [w_R \tilde{\rho}(\theta) + w_S \tilde{S}_r(\theta)], \quad (6)$$

where $\Theta_{\text{feasible}} = \{\theta \in \Theta : R(\theta) \geq R_{\text{base}} - \delta_R \text{ and } T_{\text{skip}}(\theta) \leq \beta \cdot T_{\text{DL}}\}$, and the nonnegative weights satisfy $w_S + w_R = 1$.

FAR of the skip-enabled system. The skip module $\mathcal{S}$ acts as a conservative gate: skipped frames are labeled negative directly, while passed frames are

processed by the same downstream detector $\mathcal{M}$ as in the baseline. Therefore, every false alarm produced by the skip-enabled system is also present in the baseline, implying

$$\text{FAR}(\theta) \leq \text{FAR}_{\text{base}}, \quad (7)$$

where $\text{FAR}(\theta)$ and FAR_{base} denote the false alarm rates of the skip-enabled and baseline systems, respectively. The primary safety risk is recall degradation caused by incorrectly skipped fire/smoke frames, which motivates enforcing the recall constraint as a hard gate prior to any candidate ranking.

Exclusion of FAR-based terms from the scoring objective. We derive an assumption-free interval for $\text{FAR}(\theta)$ in terms of $S_r(\theta)$ and the fixed baseline constant FAR_{base} , and then argue that explicitly optimizing for FAR reduction is unnecessary in the present setting.

By definition:

$$\text{FAR}(\theta) = \frac{FP(\theta)}{N_{\text{neg}}} \quad (8)$$

where N_{neg} is the total number of ground-truth negative frames and $FP(\theta)$ is the number of false positives produced. Since every skipped frame receives a negative label, false positives arise only from frames passed to $\mathcal{M}$:

$$FP(\theta) = FP_{\text{base}} - a, \quad a \geq 0 \quad (9)$$

where a is the number of skipped frames that $\mathcal{M}$ would have misclassified as positive. A skipped false positive must simultaneously be a baseline false positive and a skipped negative frame, so:

$$0 \leq a \leq \min(FP_{\text{base}}, N_{\text{skip_neg}}) \quad (10)$$

This bound requires no assumption on the relationship between the skip decisions of $\mathcal{S}$ and the outputs of $\mathcal{M}$. Substituting (10) into (9), dividing by N_{neg} , and using $\text{FAR}_{\text{base}} = FP_{\text{base}}/N_{\text{neg}}$ and $S_r(\theta) = N_{\text{skip_neg}}/N_{\text{neg}}$ gives:

$$\text{FAR}(\theta) \in \left[\max(0, \text{FAR}_{\text{base}} - S_r(\theta)), \text{FAR}_{\text{base}} \right] \quad (11)$$

Justification for excluding FAR optimization. Equation (11) does not imply that $\text{FAR}(\theta)$ is uniquely determined by $S_r(\theta)$: two configurations with the same skip rate may skip different frames and therefore yield different realized values of a , leading to different FAR values within the same interval. However, explicit FAR optimization is not necessary here for two reasons.

First, (11) provides a hard guarantee that $\text{FAR}(\theta) \leq \text{FAR}_{\text{base}}$ for every θ . Thus, the skip module cannot increase the false alarm rate relative to the baseline.

Second, $\mathcal{S}$ makes skip decisions using only inter-frame motion, without access to the outputs or error behavior of $\mathcal{M}$. Therefore, the hyperparameters of $\mathcal{S}$ do not directly control which baseline false positives are removed. Any observed FAR reduction depends on whether the motion patterns selected by $\mathcal{S}$ happen to overlap with frames that $\mathcal{M}$ would misclassify on the validation set. Including

FAR in the objective could therefore favor configurations that match validation-specific error patterns, which may not generalize well.

For these reasons, we exclude FAR-based terms from the scoring objective and report $\text{FAR}(\theta)$ only as an observed consequence metric in the results tables.